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The Editors
Biometrical Journal

Dear Editors,

I am pleased to submit the manuscript “**Are Halachic Vessel Patterns Overrepresented in Empirical Menstrual Cycle Data? A Permutation Test**” for consideration as a Research Article in *Biometrical Journal*.

What this paper does. Jewish religious law (*Halacha*) prescribes that women anticipate menstruation based on recurring patterns in their cycle history, collectively known as *vesset*. Despite their centuries-long practical application, no published study has formally tested whether these patterns occur in real menstrual cycle data at rates exceeding chance. This paper fills that gap using three complementary null models applied to a publicly archived dataset of 1,554 cycles from 118 women (Fehring et al., *Contraception*, 2013): a global permutation null (H_G), a multinomial sampling null (H_{iid}), and — as the primary scientific test — a within-woman permutation null (H_W) that conditions on each woman’s observed cycle-length multiset.

Methodological contributions. The paper makes three methodological contributions:

1. **A permutation-based Mahalanobis joint test for dependent count vectors.** We introduce a distribution-free joint test (D_M) that evaluates whether a vector of sequentially computed pattern counts departs from the null distribution without assuming multivariate normality or independence among components. The test is applicable to any setting where multiple sequential pattern types are counted in time-series or longitudinal biological data.
2. **A within-woman permutation null as a design-adapted exchangeability block.** H_W permutes each woman’s cycle-length sequence within her own record, conditioning on her personal cycle-length multiset. This block-randomization design controls for individual variation in cycle regularity and tests only whether temporal patterns exceed within-person chance levels — a more targeted scientific question than the global null.
3. **A polynomial hierarchy of pattern types.** The five vessel pattern types form a hierarchy based on the order of the finite-difference operator. We derive three mathematically necessary dependence relationships (set-theoretic inclusion, limiting convergence, logical

incompatibility) and confirm each with pairwise Spearman correlation tests, motivating the joint test.

Main findings. The primary result is that no vessel pattern is individually or jointly significant under the within-woman null (H_W): all Holm-corrected individual $p \geq .198$ and the joint Mahalanobis test is non-significant ($D_M = 2.95$, $p = .121$). This shows that observed pattern counts are consistent with within-person randomness once each woman's personal cycle-length distribution is controlled for, and provides the first formal permutation-based test of halachic vessel patterns in empirical data.

Under the global permutation null (H_G), Haflaga departs significantly from chance ($z = 4.51$, $p < .001$; only 2 of 50,000 simulated datasets exceeded the observed count), and the primary 5-pattern Mahalanobis joint test detects significant multivariate departure ($D_M = 5.10$, $p < .001$). The contrast between H_G and H_W is itself informative: Haflaga overrepresentation relative to the population pool reflects partly the composition of individual women's cycle-length distributions, not purely within-sequence patterning.

Fit with *Biometrical Journal*. This paper fits *Biometrical Journal*'s scope as a methodological contribution motivated by a real biological problem. The central development — a permutation-based Mahalanobis joint test for correlated count vectors arising from sequential pattern detection — is a distribution-free alternative to Hotelling's T^2 with broad applicability to cyclic biological settings (circadian rhythms, hormonal cycle data, episodic clinical event sequences). The within-woman permutation null exemplifies design-adapted exchangeability-block randomization (following Winkler et al., 2014; Nichols et al., 2002) in a novel longitudinal context. The menstrual cycle application provides concrete, publicly archived data for reproducibility.

Required declarations.

Author concurrence: The sole author concurs with this submission.

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Prior submission: This manuscript has not been submitted elsewhere and is not under review at any other journal.

Ethics: This study is a secondary analysis of a publicly archived, fully anonymised dataset (Fehring et al., 2013, archived at Marquette University). IRB approval was obtained for the original data collection by the original investigators. No new data were collected and there was no direct patient contact in this study.

Conflicts of interest: The author declares no conflicts of interest.

Closely related work: A preliminary analysis appears in the author's PhD dissertation (Ariel University, 2022, in Hebrew). The dissertation examined vessel patterns descriptively; the present paper is a substantial extension introducing the joint multivariate test, the multinomial null model, the polynomial hierarchy with mathematical derivations, the within-woman permutation null as primary test, and the pairwise Spearman dependence analysis, none of which appeared in the dissertation. The dissertation is submitted as supplementary material for review only.

Data and code availability. All analysis code and the filtered dataset are publicly available at <https://github.com/dvirross/VesselVectorTest>. The original dataset is

archived at Marquette University (https://epublications.marquette.edu/data_nfp/7/).

I believe this manuscript will be of genuine interest to the readership of *Biometrical Journal*, and I look forward to the editorial team's consideration.

Yours sincerely,

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Declarations

Funding

This research received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors.

Conflict of interest

The author declares no conflicts of interest.

Ethics approval

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Patient consent

Not applicable. No new data were collected. The original dataset was collected under the ethical oversight of the original investigators (Fehring et al., 2013).

Data availability

The filtered dataset and all analysis code used in this study are publicly available at <https://github.com/dvirross/VesetVectorTest>. The original NFP dataset is archived at Marquette University: https://epublications.marquette.edu/data_nfp/7/.

Author contributions

Dvir Ross: Conceptualization, Methodology, Software, Formal analysis, Investigation, Writing — original draft, Writing — review & editing, Visualization.

Artificial intelligence use disclosure

Portions of the written text in this article were drafted and edited with the assistance of Claude (Anthropic, 2024), a large language model AI assistant. All scientific content, mathematical derivations, statistical analyses, and intellectual contributions are solely the work of the author. The AI tool was used exclusively for language editing, structural organisation of the manuscript, and generation of $\text{\LaTeX}$ and `Python` code under the author's direct supervision and verification. The author takes full responsibility for the accuracy and integrity of all content presented in this article.

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